CLINE AI BUILDER HACKATHON · ROUND 2

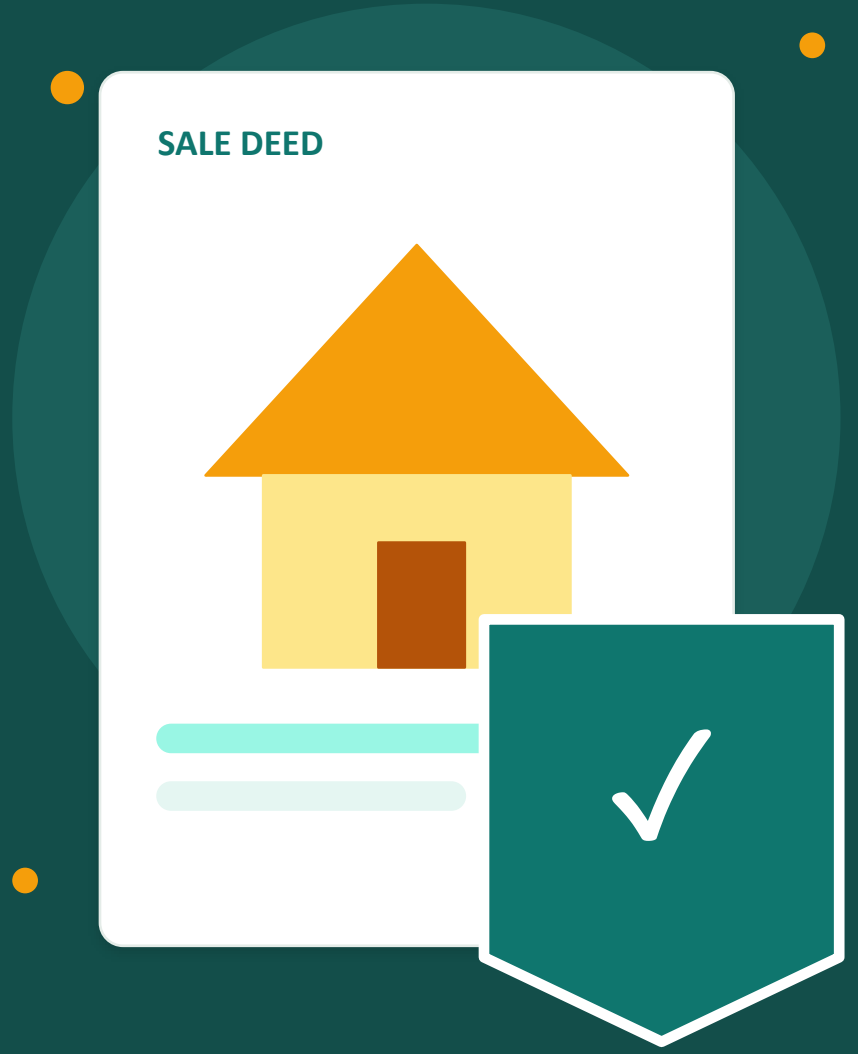
Bhoomi Kaaval

भूमि காவல்

Bhoomi = land (Sanskrit/Hindi) · Kaaval = guard (Tamil)

Check your property papers before you pay.

An AI reader plus fixed rules, with a page and a quote behind every finding. It never certifies title.



India registers deeds, not title

1

Registration is not title

Registering a deed records that it was presented and executed. It does not prove the seller owns the land, and registering officers cannot decide title.

2

Risk hides in the papers

A broken ownership chain, a second sale of the same plot, an area that does not add up: all buried in scanned deeds, in several languages.

3

No record of what was seen

When a dispute follows, there is no structured record of which risks were visible at the moment of registration.

The result: preventable property disputes, for buyers and for honest officers.

One engine, two faces

Core engine: deed bundle → structured fields → fixed rules → findings with evidence

Citizen: Check my documents

- Pick or upload deeds: PDF, scan or phone photo, English or Hindi
- See every finding with its page and quote
- Get an ask-your-seller list and an advocate brief

Sub-Registrar: Officer console

- Record a disposition and a reason for every flag
- Class B advisories can never be a refusal ground
- The decision is signed into a tamper-evident log

Design rule: the AI only reads. Fixed, explainable rules decide. Legal wording never comes from free AI text.

How it works, end to end

1 — Upload

PDF, scan or photo; SHA-256 of the original file

2 — Redact

Aadhaar and PAN masked before any image is sent

3 — AI reads

Vision model returns value, status, confidence, quote

4 — Cross-check

Text layer or Tesseract confirms the quote is on the page

5 — Rules

Class A and Class B checks, no AI involved

6 — Decide

Officer records a reason for every flag

7 — Sign + log

Hash-chained, Ed25519-signed ledger entry

Extraction: reads and verifies the page | **Deterministic** | **Human + ledger**

Anything the AI cannot read becomes “needs review”. No rule ever fires on a field we could not read, so a bad scan never produces a false accusation.

Three states, never a guess

Extracted

Read with enough confidence and a quote found on the page.

The rules use the value.

Absent

The page is readable and the item is genuinely not stated.

Class A rules may fire, for example a power of attorney with no registration number.

Unreadable

Probably there, but we cannot read it reliably.

Treated as unknown: no flag is raised and a human reviews it.

Confidence, from three signals

40% model's own score
30% value has the right format
30% quote really found on the page

Real result from our tests

On a poor scan, “POA number not stated” was not trusted. It became “needs review” instead of a false hard stop.

Class A and Class B checks

Class A · Registrability

Objective checks that may justify refusal or return for correction.

- A1 Power of attorney has no registration number
- A2 Property area and unit stated
- A3 All four boundaries described
- A4 Presented within the allowed window (draft rule)
- A5 Seller and buyer names present
- A6 Consideration (price) stated
- A7 ULPIN consistent with the parent deed

Class B · Title advisories

Signals to investigate. Never a ground for refusal.

- B0 Registry cross-check could not run
- B1 Seller is not the registry's current owner
- B2 Possible double sale in the registry index
- B3 Area exceeds the parent deed
- B4 Boundary differs from the parent deed
- B5 Dates out of order
- B6 Possible same person, spelled differently
- B7 Unregistered instrument in the chain
- B8 Encumbrance recorded in the registry

A chain gap explained by a succession document raises no B1 (our decoy case). A mock registry index stands in for NGDRS and land-stack APIs. Rules are heuristics, not legal conclusions.

Every finding shows its source

A1

Power of attorney has no registration number

🛑 **Class A · Registrability**

POA_REGISTRATION_MISSING

📍 **Page 1**

“…acting through his duly constituted attorney Anil…”

Why it matters: the deed was executed through an attorney, but gives no registration number for that power of attorney.

Next step: obtain the registered power of attorney and its registration number.

Example from a synthetic sample deed.

1

Check my documents

Findings, fields read with confidence, ask-your-seller list, advocate brief

2

Officer console

A disposition and a reason for every flag, then a signed receipt

3

Decision ledger

Chain view, one-click verify, and a tamper test on a copy

Safe by design, provable by log

Safe by design

- Aadhaar and PAN numbers are masked before any image is sent to a model
- Never “clean”, never “safe”, never a title score
- Legal wording comes from a fixed library, not AI text
- Every result lists what we did not check

Names are stored in the log only as salted commitments.

Tamper-evident decision record



Verification fails at block 3

Each entry binds: SHA-256 of the original files, extraction hash, ruleset and model version, salted name commitments, and an Ed25519 signature. A hash-chained, signed log: not a blockchain, and it proves integrity, not fault.

Models and tools

Vision LLM

Gemini Flash reads the page images. One provider interface: Claude also supported, a self-hosted model can be swapped in.

Local reader

Tesseract 5 (English, Hindi, Marathi, Assamese packs) and the PDF text layer. Cross-checks the model and finds ID numbers to mask.

Cline SDK agent

@cline/sdk on Node 22 reviews fields the extractor could not read. It can only propose; Python re-checks.

Python core

PyMuPDF, OpenCV, pydantic and rapidfuzz for ingest, preprocessing, validation and matching.

Rules + ledger

Deterministic Python rules, SQLite, SHA-256 hash chain, Ed25519 signatures, salted HMAC commitments.

Streamlit app

Website-style interface with SVG illustrations. Runs offline from cached extractions for the demo.

How Cline was used

In the build

- Cline (ClinePass, in VS Code) made real changes: a prompt fix so the extractor must return every field, plus a version bump and a test
- Cline wrote the request-list test suite and the README write-up
- Each change is a visible diff, a green test run and a commit

To be plain: the first version of the pipeline, rules bridge and ledger signing was drafted with other AI coding assistants. The commit history shows who contributed what.

Inside the product

Cline SDK review agent

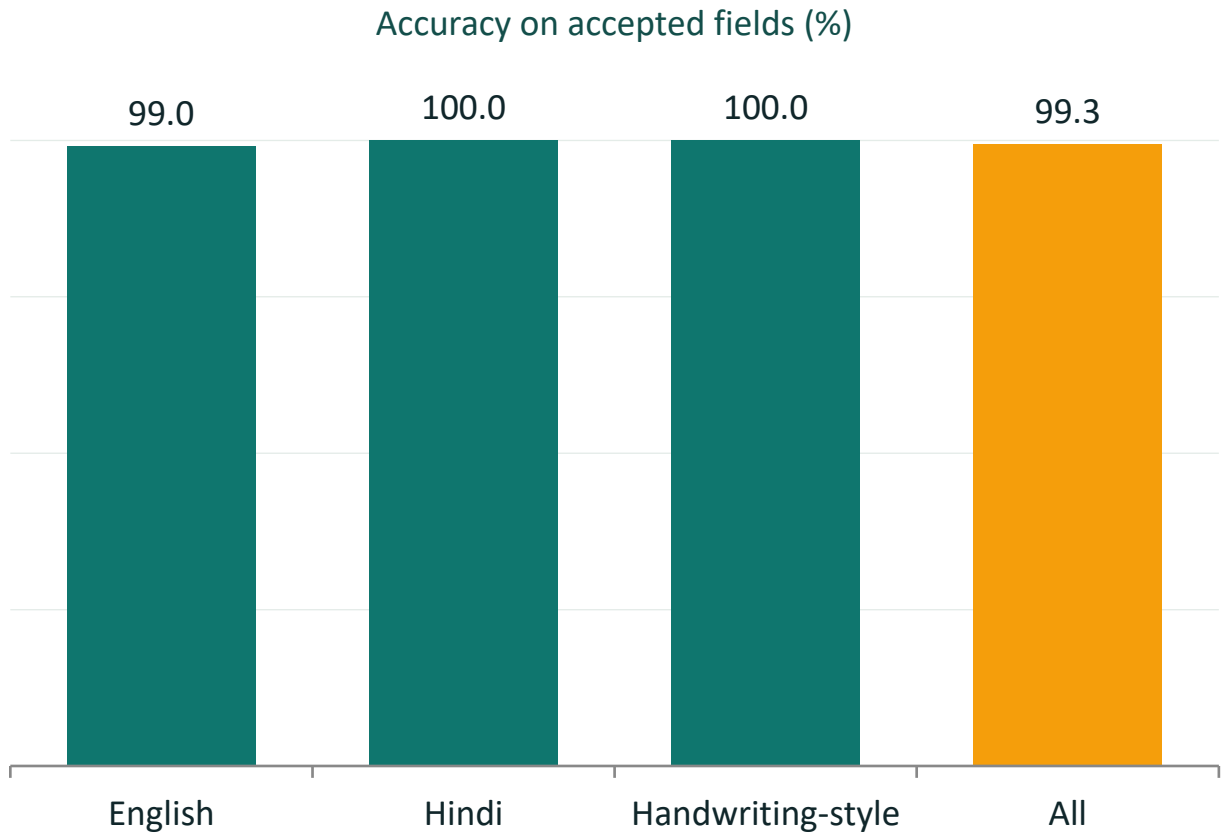
`get_field` `read_page_text`

`validate_field` `propose_correction`

Python re-checks: format valid AND quote found on the page

It has no tool to run rules, create flags or touch the ledger. It works with the agent switched off. On our test set it recovered 0 of 1 field: deliberately conservative.

Measured results



161 labelled fields, 7 synthetic deeds: clean and scanned English, Hindi, handwriting-style

11 / 11 rule cases correct on planted defects, including the succession decoy

75 offline tests pass

Synthetic data we authored; one model. One blurred “Bora” was read as “Boro”.

Rule results use canned extraction replies, so they test the rules, not OCR. Abstained fields: 1.9% (sent to review).

Challenges we hit and fixed

What went wrong	What we did
An Aadhaar number was not masked when a full stop (or the Hindi danda) followed it	Punctuation-aware matching and regression tests
Gemini spent 15.7k of 16.4k tokens “thinking”; the reply was cut off	Capped thinking; a cut-off reply is an error, never parsed
On empty forms the model looped, copying dotted blanks	Prompt rule, one re-ask, and only valid replies are cached
Extractor and registry disagreed on the ULPIN format	14-character IDs everywhere, short codes aliased
Free API tier: 20 requests per day per model	Raw-reply cache and offline sample data for the demo

Against the evaluation criteria

Creativity and uniqueness	A three-state reader on top of fixed rules, and a Class A / Class B split so officers are never pushed to refuse on title grounds
Innovation	“Needs review” instead of guessing; quotes checked against an independent reader; a signed, tamper-evident decision record
Real-world usefulness	A citizen pre-flight and an officer console, with an ask-your-seller list; built to sit on top of land-record systems, not replace them
Effective use of Cline	Cline in VS Code made reviewable changes with tests and commits (see the demo video and the commit history)
Usability and execution	Website-style app, plain-language wording, three clear flows, runs offline for the demo
Technical implementation	Python pipeline, 75 tests, masking before any model call, Ed25519 plus hash chain, swappable model provider
Bonus: Cline inside the product	A Cline SDK review agent that can only propose, with every proposal re-checked on the page

Limits and what comes next

Honest limits

- Synthetic documents, one model: this shows the logic, not real-world accuracy
- English and Hindi tested; Assamese not tested
- Handwriting tested with a handwriting font, which is tidier than real writing
- Handwritten ID numbers cannot be masked automatically
- A deed-only view cannot see inheritance, mutations or litigation

Future scope

- Pilot with one Sub-Registrar office on a sandbox registry feed
- Self-hosted Indic vision model, so documents never leave the department
- State-specific rule packs and an Indic handwriting test set from consented specimens
- Transliteration-aware name matching across scripts
- A Cline-powered assistant that answers “why was this flagged?” from the evidence

Bhoomi Kaaval

Land guarded by evidence, not guesses.

It shows what your documents support, what they contradict, and what nobody has checked.
It never certifies title.

Demo video · GitHub repository and README · Built with Cline